

CMPE 214 : Lab 01

Group 06

1) What is the CUDA driver version?

13.0 / 12.8

2) What is the CUDA compute capability?

7.5

3) What is the number of multiprocessors?

40

4) What is the number of CUDA cores per multiprocessor?

128

5) What is the size of constant memory?

65536 bytes

6) What is the max size of shared memory per thread block?

49152 bytes

7) What is the max register number per thread block?

65536

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Device 0: "Tesla T4"
  CUDA Driver Version / Runtime Version      13.0 / 12.8
  CUDA Capability Major/Minor version number: 7.5
  Total amount of global memory:             14913 MBytes (15637086208 bytes)
MapSMtoCores for SM 7.5 is undefined. Default to use 128 Cores/SM
MapSMtoCores for SM 7.5 is undefined. Default to use 128 Cores/SM
  (40) Multiprocessors, (128) CUDA Cores/MP: 5120 CUDA Cores
  GPU Max Clock rate:                        1590 MHz (1.59 GHz)
  Memory Clock rate:                         5001 Mhz
  Memory Bus Width:                           256-bit
  L2 Cache Size:                             4194304 bytes
  Maximum Texture Dimension Size (x,y,z)     1D=(131072), 2D=(131072, 65536), 3D=(16384, 16384, 16384)
  Maximum Layered 1D Texture Size, (num) layers 1D=(32768), 2048 layers
  Maximum Layered 2D Texture Size, (num) layers 2D=(32768, 32768), 2048 layers
  Total amount of constant memory:            65536 bytes
  Total amount of shared memory per block:    49152 bytes
  Total number of registers available per block: 65536
  Warp size:                                 32
  Maximum number of threads per multiprocessor: 1024
  Maximum number of threads per block:        1024
  Max dimension size of a thread block (x,y,z): (1024, 1024, 64)
  Max dimension size of a grid size (x,y,z): (2147483647, 65535, 65535)
  Maximum memory pitch:                       2147483647 bytes
  Texture alignment:                           512 bytes
  Concurrent copy and kernel execution:       Yes with 3 copy engine(s)
  Run time limit on kernels:                   No
  Integrated GPU sharing Host Memory:          No
  Support host page-locked memory mapping:     Yes
  Alignment requirement for Surfaces:          Yes
  Device has ECC support:                      Enabled
  Device supports Unified Addressing (UVA):    Yes
  Device PCI Domain ID / Bus ID / location ID: 0 / 0 / 4
  Compute Mode:
    < Default (multiple host threads can use ::cudaSetDevice() with device simultaneously) >

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Nvcc version

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/content# nvcc --version
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2025 NVIDIA Corporation
Built on Fri_Feb_21_20:23:50_PST_2025
Cuda compilation tools, release 12.8, V12.8.93
Build cuda_12.8.r12.8/compiler.35583870_0

```

OS:

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/content# cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
/content# uname -a
Linux 972f0089fc0f 6.6.122+ #1 SMP Thu Apr 30 18:17:14 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux

```